

Next-Gen Displays Enabling Super-Friendly Devices

Interactivity is emerging as a key differentiating factor in the display world, and so are other aspects such as greenness, flexibility, performance, energy consumption and heat dissipation. Sitting in the background, powerful chips and software code are working magic to achieve these and other targets



With pinches and tickles, new touch-enabled display technologies are making devices more personal than ever before (Image courtesy: www.ciccareadesign.com)

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The most conspicuously visible aspect that excites most customers about today's electronics products is the display. Be it a new cell phone, tablet, television or laptop, the first thing that people look at is how bright and clear the display is. Understandably, the displays are fast evolving. Even as older screens are giving way to high-definition (HD) ones,

quad HD and ultra HD are already on their way. Whether you are a fan of liquid-crystal displays (LCDs), light-emitting diode (LED) displays or Plasma screens, there is some improvement happening every day in their performance or clarity.

That said, the current generation of mobile users has triggered a fresh set of parameters for evaluating displays, and a brand new set of challenges for display makers and researchers. Interactivity has emerged as a key requirement, giv-

ing rise to improved touch screens and immersive technologies. The need to conserve battery life has made engineers focus their energy on reducing the power consumed by displays.

As devices are getting packed into smaller footprints, there is also a need to make cooler displays that do not affect the performance of surrounding components. People spending a lot of time outdoors love to watch movies or play games on their mobile gadgets in their free time. This is enforcing the need for higher display performance even in smaller screens.

Likewise, the need to use mobile phones in rugged outdoor environments is pushing manufacturers to come up with displays that show clear images even in bright daylight, robust and flexible displays that are not damaged by a fall, and so on. On top of this, there is the much-hyped 3D trend!

It takes a multi-disciplinary approach to solve these problems and create a perfect experience for the users. Materials sciences, nanotechnology, chemical engineering, physics, electronics, integrated circuit (IC) design and software development are all waltzing together to create the super displays we see today. Much of the magic is, in fact, done by the powerful little chips and code that work in the background. Here is a quick round-up of some interesting display-related developments in the year gone by.